Does Remote Sensing Beat the Trail Map? Predicting Hiking Speed in Unseen Regions from Lidar Terrain and OpenStreetMap Tags

Brian Zhou [TODO: author list and order: ask Danni, Tianyi and Prof. Jacobs]
Department of Computer Science and Engineering, Washington University in St. Louis

***Abstract*—[TODO: Draft by Claude: rewrite in your own words.] Estimates of walking time on trails still rely on Tobler's 1993 hiking function. We ask what corrects it in regions a model has never seen, using 290,640 walking traversals of 50 m trail segments from public OpenStreetMap (OSM) GPS traces in 17 North American mountain regions. Holding out one region at a time and scoring calibrated predictive distributions (CRPS), we find Tobler's function 12.5% worse than a refitted slope curve. Terrain from elevation models, mostly airborne lidar, improves CRPS by 2.1–2.8% in unseen regions and beats OSM trail tags in 16 of 17 (additive models); tags add 0.4–0.9%, lidar vegetation adds little, and satellite embeddings help mainly where the model was trained. For the same hiker on the same trail, descending is 8–23% faster than climbing, not 42% as Tobler implies. A terrain-adjusted hiking function with one elevation-derived input keeps part of the gain, but a hiker's own pace still matters most over whole hikes.**

***Index Terms*—hiking speed, lidar, OpenStreetMap, GPS trajectories, transfer, calibration**

I. Introduction

[TODO: Draft by Claude: rewrite in your own words.]

How long a walk will take matters for search and rescue, for wildland firefighters' escape routes [1] and for the times trail apps show. A common starting point is Tobler's hiking function [2], $v = 6\exp(-3.5\,|s + 0.05|)$ km/h on slope s, which peaks at 6 km/h on a gentle descent. It is still in use, even for traverse planning in planetary exploration [3].

Crowdsourced GPS tracks show that the function is too fast and too peaked, and that speeds at one slope vary widely [4], [5]. Newer models add covariates: OSM terrain classes and lidar-measured obstruction in England and Wales [6]; boosted trees on 16 factors, weather included, in Hong Kong [7]; a latent walking speed predicted from Sentinel-2 imagery in one Croatian range [8]; the hiker's pace on the first part of a hike [9]. STRIDE [10] models walking speed from lidar slope, understory density [11] and ground roughness, calibrated on timed field walks. On roads, speeds and their spread have been learned from overhead imagery and lidar [12], [13].

Wood et al. [6] test with 10-fold cross-validation and Huang et al. [7] with a random split, so their test tracks come from the areas they trained on. Yet predictions matter most where nobody has recorded a hike: in the White Mountains, one of our best-sampled regions, 77% of the 3,369 km of mapped trail has no trace in our data. Two sources could describe such trails. On the trail map, OSM mappers tag difficulty (sac_scale), surface and visibility, but unevenly: sac_scale is on 1–96% of walked trail, depending on the region. Remote sensing reaches every trail, tagged or not: the USGS 3D Elevation Program collects airborne lidar across the United States [14], and satellite embeddings summarize imagery worldwide [15]. To our knowledge, no study has tested which of the two works in regions a model has never seen.

We test it on public OpenStreetMap GPS traces [16] from 17 mountain regions of the United States and Canada. We contribute (i) a dataset of 290,640 walking traversals of 50 m trail segments (1,992 outings) with slope, OSM tags and remote-sensing covariates (Sec. II); (ii) a transfer test that holds out whole regions, zones or half the continent and scores predictive distributions with the continuous ranked probability score (CRPS) [17], against published hiking functions (Sec. III); and (iii) the finding that terrain measured from elevation data transfers ($+2.1$ to $+2.8\%$ CRPS skill over a refitted slope curve in unseen regions, ahead of tags in 16 of 17 with additive models), while tags add $+0.4$ to $+0.9\%$ and neither vegetation, from 30 m maps or lidar point clouds, nor satellite embeddings match it (Sec. IV). We also give a terrain-adjusted hiking function with one DEM-derived input ($+1.0\%$).

II. Data

[TODO: Draft by Claude: rewrite in your own words.]

Sources. We use public GPS traces uploaded to OpenStreetMap (OSM) [16] in 17 mountain regions of the United States and Canada (Fig. 1), chosen for relief, trail density and lidar coverage. The traces come from three sources: the OSM trackpoints API, which returns the timestamped points of identifiable and trackable uploads tile by tile; the original GPX files of identifiable uploads, which give whole hikes at the device's own sampling; and our lab's crawl of 1.5 million original files of public and identifiable uploads from 2016–2022, which keeps the timestamps the API drops. Sixteen regions combine all three; the Bay Area uses the crawl only, because the API times out on its tiles.

Processing. Each recording is cleaned (repeated timestamps, steps faster than 60 km/h), split at gaps over 5 min and thinned to points at least 5 s apart. A step counts as a stop when the point 60 s later lies within 10 m. Traces are map-matched to the OSM trail and road network with a hidden Markov model [18] (85–99% of points matched, by region), and the matched path is cut into 50 m segments.

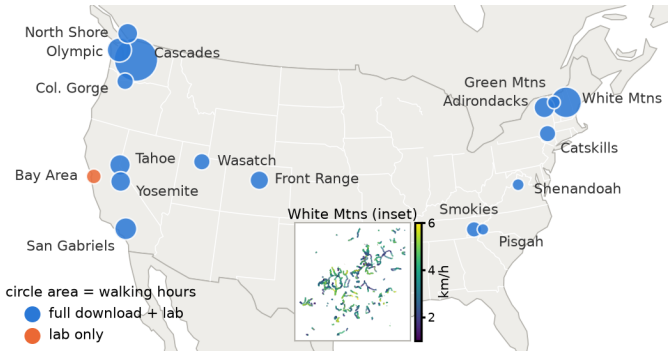


Fig. 1. The 17 regions. Circle area is walking hours (1,992 outings, 4,702 h in total). Blue: full OSM download plus the lab crawl; orange: lab crawl only. Inset: the White Mountains' 34,535 walking segments colored by speed.

Each segment records its moving speed, its signed slope along the direction of travel from the 5 m USGS 3DEP elevation grid [14] (resampled from 1 m lidar under at least 83% of walked trail in 13 regions), the OSM tags of its way and the trace's recording interval. Duplicate uploads are removed, and traces that start within 15 min and 1 km of each other form one outing (a group hike), the unit we resample. Travel mode is inferred per trace, anchored on trails closed to bikes and horses. Against the activity labels in 327 traces' files, 93% of labelled hikes are classed as walking, and 5% of labelled traces we class as walking say bike or ski.

The dataset. The regions hold 2,876 hiking traces. The walking analysis set (walking traces on trails) has 290,640 segments (14,532 km) from 2,046 traces in 1,992 outings and 4,702 moving hours. Its uploaders are OSM mappers and enthusiasts rather than a random sample of hikers. Sparse traces read slow, because straight lines between points cut corners and short stops hide between them: thinning dense walks to 30 s between points lowers their measured speed by 11%. The gains we report hold at every recording rate (Sec. IV).

III. Methods

[TODO: Draft by Claude: rewrite in your own words.]

Target and covariates. We predict the distribution of $\ln v$, the log moving speed of a walking segment (289,679 with grades within $\pm 60\%$). Covariates come in six groups: (a) slope; (b) OSM tags: sac_scale, trail_visibility, surface, width, smoothness, mtb_scale, highway type and permitted modes; (c) terrain from elevation, read every 1 m along the segment's matched path: the spread (SD) and maximum of its 5 m pitches, the relief around an 11 m moving average and the side slope, from the 5 m grid and, where available, 1 m lidar, which adds STRIDE's ground roughness [10]; (d) 2021 land cover and tree canopy at 30 m [19]; (e) 2022 AlphaEarth satellite embeddings at the segment midpoint [15], [20]; (f) in the 15 regions with lidar point clouds, understory density (returns at 0.85–1.2 m among those below 1.2 m [11]), canopy cover and height within 3 m of the path. As a control, "hidden climbing" is the extra time Tobler's function predicts when the segment is walked as ten 5 m pieces instead of at its average grade.

Models. The baselines are Tobler's function [2], the published median curve of Campbell et al. [5], STRIDE [10] as published (understory density set to 0, as most regions lack point clouds) and with its form refitted, and our reference: a rounded, asymmetric slope curve refitted to the training data, $\ln v = a - k_{\pm}\left(\sqrt{(s+s_0)^2+e^2}-e\right)$, with k_+ above the peak and k_- below it. Each covariate group then enters two models: an additive one (ridge regression of the curve's residuals on one-hot bins: deciles of continuous covariates, one level per tag value plus "missing") and gradient-boosted trees on slope and the covariates. The terrain-adjusted hiking function keeps Tobler's form but lets peak speed and slope sensitivity depend on one DEM-derived number, the steppiness z:

$$\ln v = a_0 + a_1 z - (k_0 + k_1 z)\,|s + s_0|, \tag{1}$$

with z the SD of the path's 5 m pitches divided by 0.10 and v in km/h; on all walking segments, $a_0 = 1.598$, $a_1 = -0.173$, $k_0 = 2.152$, $k_1 = -0.392$ and $s_0 = 0.043$.

Predictive distributions and tests. Every model predicts its point estimate plus the quantiles of its residuals on held-out outings of the training data (training outings split 70/15/15 into fitting, early stopping and calibration), so calibration has to transfer too. Splits never separate a segment from its hike: 5 folds of outings (new hikes; a hiker's other outings may be in training), 5 folds of 10 km UTM squares (new places) and, the headline, leave one region out (17 folds); stricter checks hold out whole zones or half the continent. We report the CRPS [17] of $\ln v$ as skill against the refitted slope curve, $1 - \mathrm{CRPS}/\mathrm{CRPS}_{\mathrm{slope}}$, with 95% intervals that resample outings (whole regions for unseen regions); the coverage of 90% intervals; and the median error in moving time over whole hikes after their first kilometre. A reference shows how much is the hiker: the median residual of a hike's first kilometre, shrunk by 0.61 (fitted on calibration outings), applied to the rest. Asymmetry is the ratio of speed down to speed up on out-and-back pairs (one outing, one OSM way, mean grade at least 3%).

IV. Results

[TODO: Draft by Claude: rewrite in your own words.]

Tobler's function fails everywhere. Walking speed peaks near 4.2 km/h on level to gently descending trail, not at 6 km/h (Fig. 2a): Tobler's function is about 20% too fast within $\pm 10\%$ grade and 18% too slow on climbs steeper than 25%, and in unseen regions it scores -12.5% (Table I), worse in all 17. Campbell et al.'s published median curve [5] is about 20% too fast on climbs steeper than 25%; it ties our refit on average (-0.5%) but ranges by region from -3.8% (White Mountains) to $+5.8\%$ (Shenandoah). Pooled over unseen regions, every model's 90% intervals cover about 89% of segments (as little as 81% in the North Shore).

Downhill is faster, but less than Tobler implies. For the same hiker on the same trail (2,763 out-and-back pairs from 1,145 outings; Fig. 2b), descending is 8% faster than climbing

TABLE I
CRPS SKILL (%) VS THE REFITTED SLOPE CURVE: ADDITIVE / TREES; STRIDE AS PUBLISHED (REFITTED).

Model	New hikes	New 10 km	**Unseen region**
Tobler 1993	−12.8	−12.7	−12.5
Campbell 2022	−0.9	−0.9	−0.5
STRIDE 2024	−17.2 (+0.2)	−17.3 (+0.2)	−16.8 (+0.2)
OSM tags	+1.3 / +1.5	+1.1 / +1.5	+0.9 / +0.4
30 m vegetation	+0.0 / +0.1	−0.0 / −0.1	−0.1 / −0.8
Embeddings	+1.4 / +2.6	+1.4 / +1.9	+1.1 / +1.1
Lidar vegetation†	+0.4 / +0.5	+0.4 / +0.3	+0.3 / +0.2
Terrain	+2.2 / +3.2	+2.1 / +2.8	**+2.1 / +2.8**
Terrain + emb. + tags	+3.7 / +5.1	+3.8 / +5.1	+3.0 / +3.9
Eq. (1), one input	+1.0	+1.0	+1.0

†The 15 regions with lidar point clouds.

at 3–10% grade and 22–23% above 20%, not the 42% Tobler implies. Our pooled curve implies nearly the same ratios, so its asymmetry is not an artifact of who walks which slopes.

Map tags transfer less well. OSM tags improve on the slope curve for new hikes in known regions (+1.3% / +1.5%, additive / trees). In unseen regions the additive model keeps two thirds of that (+0.9% [0.1, 1.7]) and the trees lose most of it (+0.4% [−1.1, 1.6]).

Terrain transfers. Terrain from elevation gains nearly as much in unseen regions as within them, +2.1% [1.5, 3.1] / +2.8% [1.8, 4.1]. It beats tags in 16 of 17 held-out regions (Fig. 3b; 14 of 17 with trees) and, on average, even a slope curve refitted on the held-out region's own other hikers (+1.2%). Terrain minus tags is +0.7 to +2.2 points (95% interval), above zero in every bootstrap draw. Uneven grade inside the segment, which one average slope hides, carries over half of the gain: the hidden-climbing control alone gives +1.2% / +1.6%. Terrain is neither a region proxy (within regions it adds ∼1.9 points to a learned offset per region) nor a hiker proxy (it adds ∼2.3 points to the hiker's own pace).

No other input transfers as well. 30 m land cover and canopy add nothing. AlphaEarth embeddings help within regions (+1.4% / +2.6%) but less, and less reliably, in unseen ones (+1.1%). Vegetation from the lidar point clouds adds at most +0.2 points to terrain, within or across the 15 regions. Terrain + tags gives +2.6% / +3.0% in unseen regions (Fig. 3a). Published STRIDE [10] scores −16.8%; refitted, +0.2%.

Robustness. The terrain gain holds for newer hikes (trained to 2019, tested on 2020 on: +2.2% / +2.9%; tags +0.9%), at every GPS recording rate (traces every ≤2 s, 2–10 s and >10 s: +1.7%, +2.4% and +1.9% in unseen regions, additive) and with no neighbouring region in training: with a whole zone held out, +1.9% / +2.7% (tags +0.7% / −0.1%); from West to East and back, +2.6% / +3.3% (tags +1.5% / −0.3%).

Whole hikes: the hiker dominates. Over 1,354 hike traces in held-out regions, the median error in moving time after the first kilometre is 13.6% for the slope curve and 13.0% with terrain, but 10.0% with the hiker's pace on that kilometre and 9.6% with both.

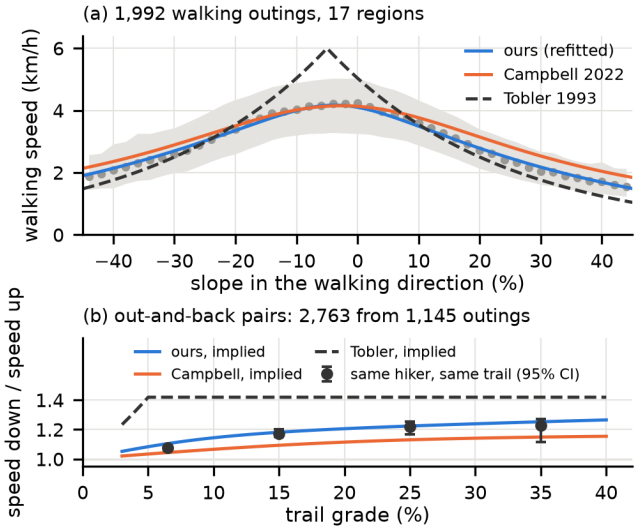


Fig. 2. (a) Walking speed vs slope in the walking direction: median per 2% bin (dots), middle 50% of segments (gray), our refitted curve, Campbell et al.'s median curve [5] and Tobler's function [2]. (b) Speed down / speed up for the same hiker on the same trail (out-and-back pairs; 95% intervals resample outings) and the ratio each curve implies.

V. DISCUSSION AND CONCLUSION

[TODO: Draft by Claude: rewrite in your own words.]

Why tags transfer less well. Not because tag values mean different things in different places: within the same hike, harder sac_scale grades are slower in every region with enough tagged trail (mountain_hiking vs hiking −5%, 10 of 10 regions; harder grades −11%, 8 of 8). Tags are mapped unevenly (sac_scale is on 1–96% of walked trail, by region), but in unseen regions the additive tag model scores about the same on tagged and untagged segments; the loss is mostly in the boosted trees (+1.5% → +0.4%), as if they fit region-specific tag combinations. Terrain exists on every trail, and its effect has the same sign in all 17 regions (steppiest vs smoothest fifth, −12%).

What terrain captures. Uneven grade within the segment, such as rock steps and short pitches, plus difficulty beyond grade. In the additive model its gain is largest on descents and level trail (+2.3 to +2.8%) and smallest on steep climbs (+0.8%), as if footing sets the pace going down and effort going up; the trees, though, gain +2.6% on steep climbs. The signal is the trail's elevation profile, not micro-relief: 1 m lidar adds little over the 5 m grid, and STRIDE's off-path roughness and understory terms add nothing on trails or make it worse.

Use. The terrain-adjusted function, Eq. (1), needs only a DEM. Fitted on the other 16 regions, it gives a pace for all 3,369 km of White Mountains trail, 77% of which no trace in our data covers. It travels less far than the full terrain model, though (+0.6% with a whole zone held out, −0.3% between East and West).

Limits. Our hikers are OSM uploaders, not a random sample. Sparse traces read slow (Sec. II), as hiking functions

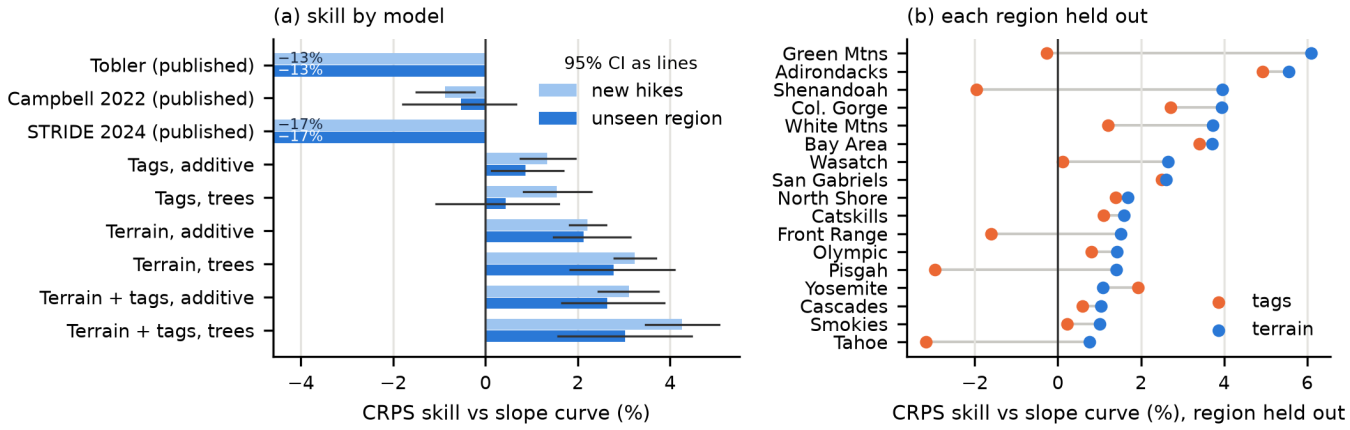


Fig. 3. CRPS skill vs the refitted slope curve. (a) By model, for new hikes in known regions (light; 95% intervals resample outings) and in regions the model never saw (dark; intervals resample regions). (b) Each region held out: terrain vs OSM tags (additive models); terrain beats tags in 16 of 17.

depend on the sampling resolution [3]; distance along the mapped trail would raise speeds by about 1–3%. Per-segment gains are a few percent because GPS noise and the hiker dominate. Weather [21] (a factor in [7]) slows our hikers on hot days (-3% above 25°C) and with snow cover (-4%), but neither weather (-0.0%) nor date and time of day improves predictions in new regions, and once a hiker's pace on the day is known, weather adds nothing.

Conclusion. Across 17 North American regions, terrain from elevation data predicts hiking speed in regions a model has never seen better than OSM tags, which keep only part of their gain there; vegetation adds little, and satellite embeddings help mainly where the model was trained. Tobler's function is too fast on gentle trail and overstates the speed-up downhill. The terrain-adjusted function keeps part of the terrain gain and needs only a DEM, though a hiker's own pace matters most. [TODO: Data release (segments without user names) if the mentors agree.]

References

[1] M. J. Campbell, P. E. Dennison, and B. W. Butler, "A LiDAR-based analysis of the effects of slope, vegetation density, and ground surface roughness on travel rates for wildland firefighter escape route mapping," *International Journal of Wildland Fire*, vol. 26, no. 10, pp. 884–895, 2017.

[2] W. Tobler, "Three presentations on geographical analysis and modeling," National Center for Geographic Information and Analysis (NCGIA), Tech. Rep. 93-1, 1993.

[3] A. Goodwin, M. Hammett, and M. Harris, "The application of Tobler's hiking function in data-driven traverse modelling for planetary exploration," *Acta Astronautica*, vol. 228, pp. 265–273, 2025.

[4] M. J. Campbell, P. E. Dennison, B. W. Butler, and W. G. Page, "Using crowdsourced fitness tracker data to model the relationship between slope and travel rates," *Applied Geography*, vol. 106, pp. 93–107, 2019.

[5] M. J. Campbell, P. E. Dennison, and M. P. Thompson, "Predicting the variability in pedestrian travel rates and times using crowdsourced GPS data," *Computers, Environment and Urban Systems*, vol. 97, p. 101866, 2022.

[6] A. Wood, W. Mackaness, T. I. Simpson, and J. D. Armstrong, "Improved prediction of hiking speeds using a data driven approach," *PLOS ONE*, vol. 18, no. 12, p. e0295848, 2023.

[7] H. Huang, J. Y. Tsou, H. Zheng, R. Ma, H. Shi, and B. Zhang, "Modeling hiking speed via interpretable machine learning: Uncovering nonlinear impacts of multidimensional environmental factors," *Applied Geography*, vol. 192, p. 104052, 2026.

[8] L. Šerić, M. Tavra, I. Racetin, and A. Ivanda, "Modeling walkability by remote sensing as latent walking speed extracted from multiple digital trail maps," *Journal of Spatial Information Science*, no. 25, pp. 67–88, 2022.

[9] M. Asako, Y. Tahara, A. Ohsuga, and Y. Sei, "Deep learning-based travel time estimation in hiking with consideration of individual walking ability," *Cybernetics and Information Technologies*, vol. 24, no. 4, pp. 3–21, 2024.

[10] M. J. Campbell, S. L. Cutler, and P. E. Dennison, "A singular, broadly-applicable model for estimating on- and off-path walking travel rates using airborne lidar data," *Scientific Reports*, vol. 14, no. 1, p. 21393, 2024.

[11] M. J. Campbell, P. E. Dennison, A. T. Hudak, L. M. Parham, and B. W. Butler, "Quantifying understory vegetation density using small-footprint airborne lidar," *Remote Sensing of Environment*, vol. 215, pp. 330–342, 2018.

[12] A. Hadzic, H. Blanton, W. Song, M. Chen, S. Workman, and N. Jacobs, "RasterNet: Modeling free-flow speed using LiDAR and overhead imagery," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, 2020, pp. 826–834.

[13] S. Workman and A. Hadzic, "Probabilistic image-driven traffic modeling via remote sensing," in *European Conference on Computer Vision (ECCV)*, ser. Lecture Notes in Computer Science, 2024, pp. 433–449.

[14] L. J. Sugarbaker, D. F. Eldridge, A. L. Jason, V. Lukas, D. L. Saghy, J. M. Stoker, and D. R. Thunen, "Status of the 3D Elevation Program, 2015," U.S. Geological Survey, Open-File Report 2016-1196, 2017.

[15] C. F. Brown *et al.*, "AlphaEarth Foundations: An embedding field model for accurate and efficient global mapping from sparse label data," 2025, arXiv:2507.22291.

[16] OpenStreetMap contributors, "OpenStreetMap," 2026, GPS traces and map data, ODbL; https://www.openstreetmap.org.

[17] T. Gneiting and A. E. Raftery, "Strictly proper scoring rules, prediction, and estimation," *Journal of the American Statistical Association*, vol. 102, no. 477, pp. 359–378, 2007.

[18] P. Newson and J. Krumm, "Hidden Markov map matching through noise and sparseness," in *ACM SIGSPATIAL International Conference on Advances in Geographic Information Systems*, 2009, pp. 336–343.

[19] J. Dewitz, "National Land Cover Database (NLCD) 2021 products," 2023, data release.

[20] Google and Google DeepMind, "AlphaEarth Foundations satellite embedding dataset," 2025, annual 10 m embeddings, CC-BY 4.0.

[21] M. M. Thornton, R. Shrestha, Y. Wei, P. E. Thornton, S.-C. Kao, and B. E. Wilson, "Daymet: Daily surface weather data on a 1-km grid for North America, version 4 R1," 2022.